E-COMMERCE PLATFORM FOR PERSONALIZED RETAIL EXPERIENCES

Project Group: 25-26J-526

Problem

Modern e-commerce platforms fail to deliver inclusive, personalized, and explainable shopping experiences, especially for Sinhala-speaking users and SMEs.

- Weak Sinhala voice understanding (accents, retail terms, code-mixed speech).
- Generic promotions that ignore customer context and intent.
- Slow, opaque BI systems that SMEs cannot act on in real time.
- Static loyalty models that do not adapt to changing customer behavior.

Impact

- Customers abandon platforms due to poor relevance and usability.
- Sinhala-speaking users are digitally excluded from voice commerce.
- SMEs lose revenue due to ineffective promotions and lack of insights.
- Low trust in AI-driven recommendations due to lack of transparency.

Who is affected & Who Benefits

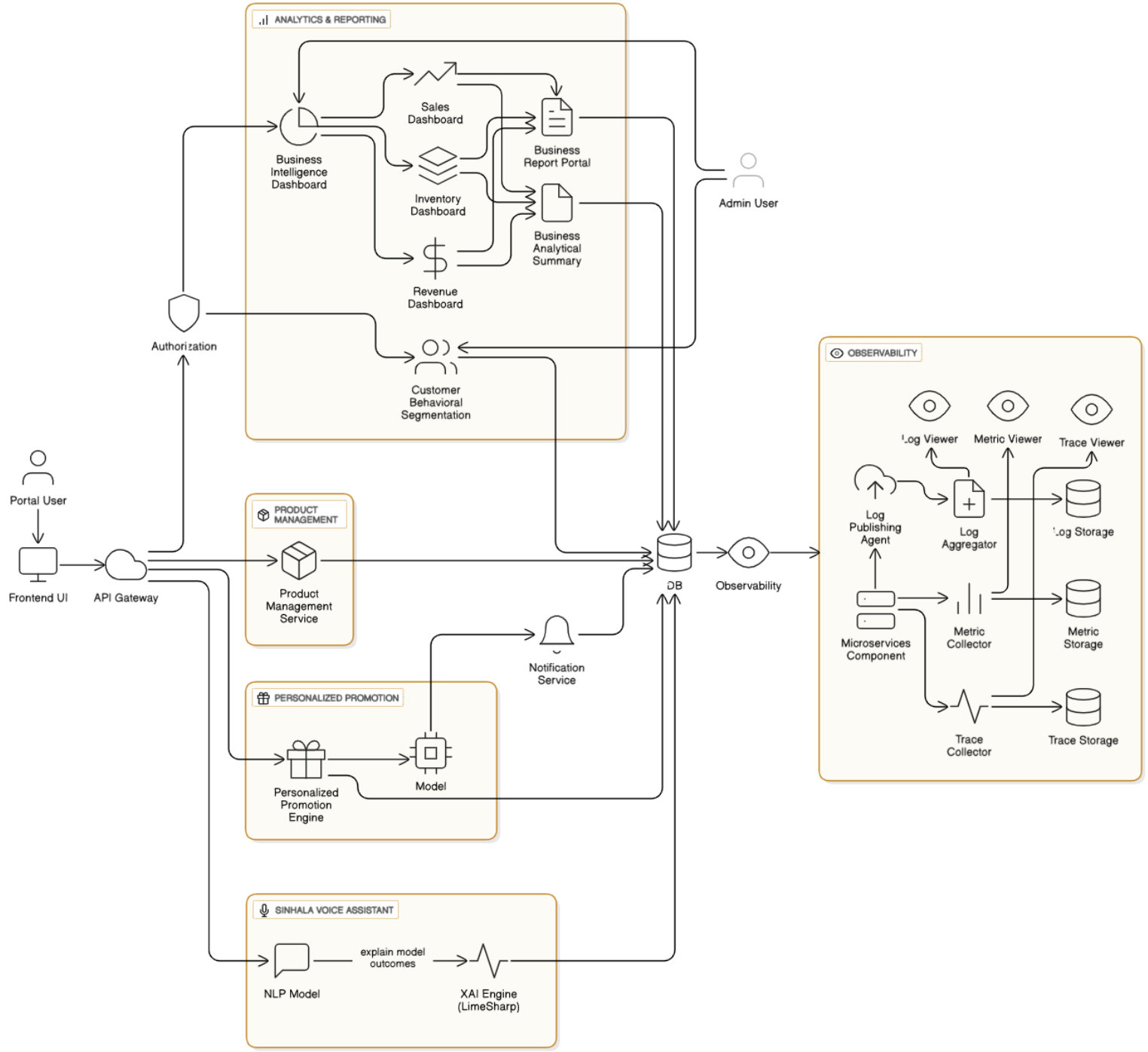
Customers
- Sinhala users
- Elderly and accessibility focused users

- Natural voice-based shopping in Sinhala
- Relevant, personalized offers
- Transparent recommendations they can trust

Small & Medium Enterprises (SMEs)
- Local retailers

- Realtime, explainable BI insights
- Smarter promotions with measurable uplift
- Adaptive loyalty systems that grow customer lifetime value

Other Benefits
- Reduced digital divide
- Increased participation of local businesses



Overall System Diagram

AI Based Personalized Promotion Engine

Specialization: Software Engineering

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08/01/2026

AI Based Personalized Promotion Engine

Identified Sub-Problems

- Bulk promotions lead to wasted discounts and low effectiveness
- No data-driven method to rank customers by purchase likelihood
- Cold-start systems cannot apply ML-based promotion logic
- Promotion systems depend on static datasets instead of live databases

Prototype User Requirements Addressed by the Solution

- Compatible with **small and large datasets**
- Admin can create **bulk or personalized promotions**
- Admin can specify **discount percentage and promotion type**
- System automatically selects **eligible customers** for personalized promotions
- Customers only see **valid promotions assigned to them**

How the Solution Addresses the Sub-Problems

The personalized promotion engine predicts customer purchase likelihood and targets only high-probability customers for personalized promotions, while using rule-based fallback strategies during cold-start conditions

User Feedback on Prototype

Initial feedback indicates that targeted promotions are more relevant to customers compared to bulk promotions, improving engagement and reducing promotion fatigue.

AI Based Personalized Promotion Engine

When a personalized promotion is created, the system evaluates customer transaction history and behavior to rank customers by purchase likelihood. Only the top eligible customers receive the promotion through the notification service.

Prototype User Requirements Addressed

- Works with **live database data**, not preprocessed files
- Compatible with **small and large datasets**
- Supports real-time promotion decision making
- Ensures secure customer-specific promotion visibility

Design Excellence / Contribution

- Separation of business logic and prediction logic
- In-memory feature generation (no storage overhead)
- Modular architecture allows future model replacement
- Aligns with real-world retail promotion systems

User Feedback on Prototype

Admin users appreciated the ability to control promotion type and discount limits, while customers preferred receiving fewer but more relevant promotions

Sinhala Voice Assistant for E-commerce (ASR + Intent + XAI)

Specialization: Software Engineering

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Sinhala Voice Assistant for E-commerce

Sub-Problems

- High error rates in **general Sinhala ASR**
- Poor handling of **retail intent & entities**
- Lack of **transparency in voice-based recommendations**
- Fragmented pipeline between ASR, intent, and personalization

End Users (Customers)

- Natural Sinhala voice interaction
- Minimal effort shopping (hands-free)
- Clear understanding of recommendations
- Fast and accurate responses

Retailers / SMEs

- Simple integration with e-commerce platforms
- Actionable customer insights
- Trustworthy AI-driven personalization
- No requirement for technical expertise

Sinhala Voice Assistant for E-commerce

Design Excellence & Key Contributions

Human-Centered Design

- Voice-first interaction tailored to Sinhala speakers
- Accessibility-oriented (elderly, low digital literacy users)

Modular & Scalable Architecture

- ASR, NLP, Recommendation, and XAI as independent services
- Easy extension to new domains and languages

Explainability as a Core Feature

- XAI embedded **by design**, not as an afterthought
- Enhances trust, adoption, and ethical AI usage

Research Contribution

- Practical application of Sinhala ASR in retail
- Demonstrates real-time voice-based XAI
- Bridges gap between low-resource language research and industry systems

Customer Behavioral Segmentation with Loyalty System Integration

Specialization: Software Engineering

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08/01/2026

Customer Behavioral Segmentation with Loyalty System Integration

Sub-Problems

- Rule-based RFM segmentation fails to detect hidden or unexpected behaviors.
- Pure RFM assumes predefined value thresholds.

Proposed Solution

- K-Means clustering as Parent Segmentation Layer.
- Groups customers based on overall behavioral similarity.
- Detects *unexpected and non-linear behavior patterns*.

Sub-Clustering with RFM

- Apply RFM analysis within each K-Means cluster.
- Creates behavior-aware sub-clusters.
- Reveals value differences inside behaviorally similar groups.

Customer Behavioral Segmentation with Loyalty System Integration

User Requirements Addressed

- Detect hidden customer behaviors.
- Differentiate value within similar behavior groups.
- Support dynamic loyalty tier creation.

Design Excellence and Contribution

- Two-layer segmentation architecture.
- K-Means captures emergent behavior patterns.
- RFM sub-clusters add interpretability and business meaning.
- Reduces bias of static segmentation models.

Business Intelligence Dashboard with XAI

Specialization: Software Engineering

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Business Intelligence Dashboard with XAI

Sub Problems

- Manual demand forecasting is inaccurate and time-consuming
- Retailers lack visibility into inventory decision reasoning

How the Solution Addresses the Sub-Problem / Prototype

- Prophet model automates demand prediction using historical sales data
- XGBoost + SHAP provides clear explanations for each forecast
- Converts predictions into restock alerts with urgency levels

Business Intelligence Dashboard with XAI

User Requirements Addressed

- Accurate daily/weekly sales forecasts
- Understandable AI reasoning (no black-box)
- Automated low-stock alerts with urgency levels

Design Excellence / Contribution

- Hybrid data pipeline (Kaggle + PostgreSQL)
- Scheduled model retraining for continuous improvement
- API-first, scalable architecture

E-Commerce Platform for Personalized Retail Experiences

Commercialization

- SaaS platform with mobile app support
- Tiered pricing packages (Basic, Pro, Enterprise)
- Enterprise licensing with customization and support

Customer Persona

- Retail shop owners (small to large)
- E-commerce platforms seeking accessibility features
- Businesses targeting rural Sri Lankan markets

Target Market & Customers

- Sri Lankan retail market: $15B+
- Target segment: 5,000+ medium to large retail stores

What is Unique in Your Solution

- First-mover in Sinhala e-commerce voice AI integration
- Bilingual XAI (Sinhala + English) for retail forecasting
- 40% improvement in rural user engagement
- Scalable for other low-resource languages

How will the cost be recovered?

- Enterprise licensing and customization fees
- Covers labor, infrastructure, AI APIs, and security

Thank You!